Diodes and Rectifiers

Q1 : explain PN junction :

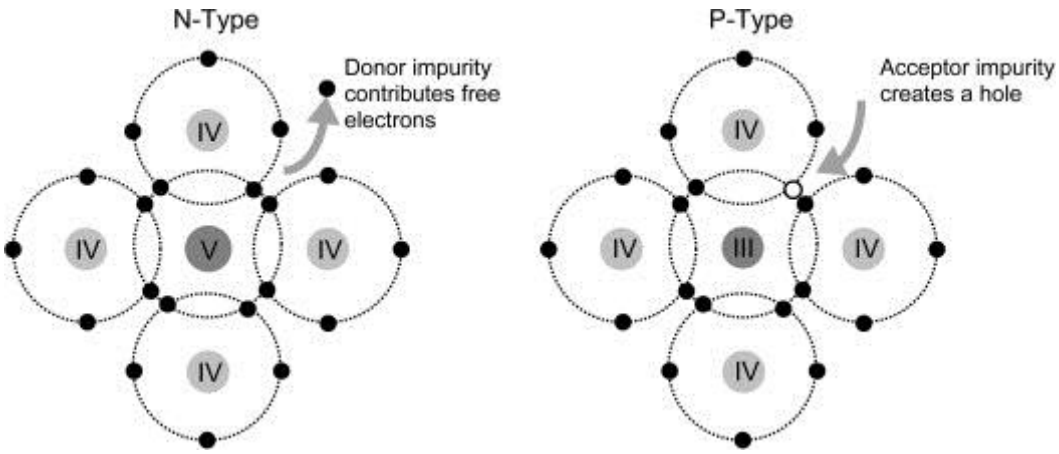
A **PN junction diode** is a fundamental electronic component that allows current to flow in one direction while blocking it in the opposite direction. It is created by joining two types of semiconductor materials: **P-type** and **N-type**. Here's a breakdown of its structure and working principle:

❖ P-TypeSemiconductors

- Siliconthathasbeendopedwithatrivalentimpuritieslikealuminum, boronandgalliumiscalled a p-type semiconductor, where the p stands for positive.
- Since holes outnumber free electrons, the holes are referred to as the majority carriers and the free electrons are known as the minority carriers.

❖ N-Type Semiconductors

- Silicon that has been doped with a trivalent impurities like arsenic, antimony and phosphorus is called an n-type semiconductor, where the n stands for negative.
- Since free electrons outnumber the holes in an n-typesemiconductor, the free electronsarecalled as the majority carriers and the holes are called as the minority carriers.



Q2 What Is forward bias for PN junction?

❖ **ForwardBias**

- Figure-6 shows a dc source across a diode. The negative source terminal is connected to the n- type material, and the positive terminal is connected to the p-type material. This connection produces what is called forward bias.

Figure-6Forwardbias

❖ FlowofFreeElectrons

- InFigure-6,thebatterypushesholesandfreeelectronstowardthejunction.
- If the battery voltage is less than the barrier potential, the free electrons do not have enough energy to get through the depletion layer.
- When they enterthedepletion layer,the ions will pushthembackintothe n region. Becauseof this, there is no current through the diode.
- When the dc voltage source is greater than the barrier potential,thebatteryagainpushesholes andfreeelectronstowardthejunction.Thistime,thefreeelectronshaveenoughenergytopass through the depletion layer and recombine with the holes.
- If youvisualizealltheholes inthep regionmoving totherightand allthefreeelectronsmoving to the left, you will have the basic idea.
- Somewhereinthevicinityofthejunction,theseoppositechargesrecombine.Sincefreeelectrons continuouslyentertherightendofthediodeandholesarebeingcontinuouslycreatedattheleft end, there is a continuous current through the diode.

  Afterthefree electron leaves the negative terminal of thebattery, it enters the right end of the diode. It travels through the n region until it reaches the junction.
- Whenthebatteryvoltageisgreaterthan0.7V,thefreeelectronhasenoughenergytogetacross the depletion layer.

- Soonafterthefreeelectronhas enteredthepregion,itrecombineswitha hole.
- Inotherwords,thefreeelectronbecomesavalenceelectron.Asavalen ce electron,itcontinues to travel to the left, passing from one hole to the next until it reaches the left end of the diode.
- Whenitleavestheleftendofthediode,anewholeappearsandtheproce ssbeginsagain.Since therearebillionsofelectrons takingthesamejourney,wegetacontinuouscurrentthroughthe diode.
- Aseriesresistorisusedtolimittheamountofforward current.

Q3 What Is reverse bias for PN junction?

- Turn the dc source around and you get Figure-7. This time, the negative battery terminal is connected to the p side and the positive battery terminal to the n side.
- Thisconnectionproduceswhatiscalledreversebias.

Figure-7Reversebias

❖ DepletionLayerWidens

- The negative battery terminal attracts the holes, and the positive battery terminal attracts the freeelectrons.Becauseofthis,holesandfreeelectronsflowaway fromthejunction.Therefore, the depletion layer gets wider.
- HowwidedoesthedepletionlayergetinFigure-8?

Whentheholesandelectronsmoveawayfromthejunction,thenewlycreatedionsincre asethe differenceofpotentialacrossthedepletionlayer.Thewiderthedepletionlayer, thegreaterthe difference of potential.

- The depletion layer stops growing when its difference of potential equals the applied reverse voltage. When this happens, electrons and holes stop moving away from the junction.

Figure-8Depletionlayer Figure-9increasingreversebiaswidensdepletionlayer

- Sometimesthedepletionlayerisshownasashadedregionlikethatof Figure-9.Thewidthofthis shaded region is proportional to the reverse voltage. As the reverse voltage increases, the depletion

layer gets wider.

❖ Minority-CarrierCurrent

- Is there any current after the depletion layer stabilizes? Yes. A small current exists with reverse bias. Recall that thermal energy continuously creates pairs of free electrons and holes. This means that a few minority carriers exist on both sides of the junction.
- Most of these recombine with the majority carriers. But those inside the depletion layer may exist long enough to get across the junction.
- Whenthishappens,asmallcurrentflowsintheexternalcircuit.
- Figure 10 illustrates the idea. Assume that thermal energy has created a free electron and hole near the junction.
- The depletion layer pushes the free electron to the right, forcing one electron to leave the right end of the crystal. The hole in the depletion layer is pushed to the left. This extra hole on the p side lets one electron enter the left end of the crystal and fall into a hole.
- Since thermal energy is continuously producing electron-hole pairs inside the depletion layer, a small continuous current flows in the external circuit.
- The reverse current caused by the thermally produced minority carriers is called the saturation current.Inequations,thesaturationcurrentissymbolizedby***IS***.Then amesaturationmeansthat we cannot get more minority-carrier current than is produced by the thermal energy
- Inotherwords,increasingthereversevoltagewillnotincreasethenu mberofthermallycreated minority carriers.

Figure-10Thermalproductionoffreeelectronandholeindepletionlayerproducesreverseminority-saturationcurrent

Q4 Explain Zener diode and its working characteristics and application

A **Zener diode** is a special type of diode designed to operate in **reverse breakdown** without damage. In reverse bias, it maintains a **constant voltage** (called the **Zener voltage**) once the reverse voltage exceeds a specific threshold, known as the **Zener breakdown voltage**.

Working:

- **Forward Bias**: Behaves like a regular diode, allowing current flow above the threshold voltage (~0.7V for silicon).
- **Reverse Bias**: When reverse voltage exceeds the Zener voltage, the diode conducts and maintains a stable voltage across it.

Applications:

1. **Voltage Regulation**: Used to maintain a stable output voltage in power supplies.
2. **Overvoltage Protection**: Protects circuits by clamping the voltage to a safe level.
3. **Clipping Circuits**: Limits the voltage in waveform shaping.
4. **Precision Voltage Reference**: Used in circuits requiring a fixed voltage reference.

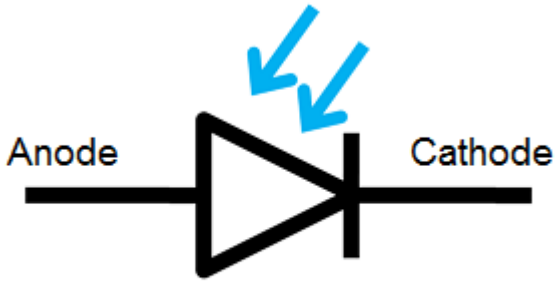
Key Characteristics:

- Precise breakdown voltage.
- Operates effectively in reverse breakdown without damage.
- Used for **voltage regulation** and **protection**.

Q5Give symbol construction, characteristics and working of photo diode:

Photodiode:

- **Symbol**: Similar to a regular diode, with arrows indicating light incidence.



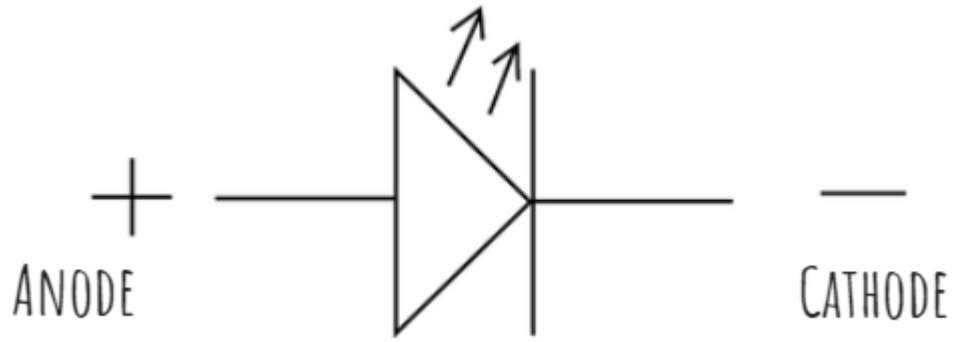
Photodiode symbol

- **Construction**: Made of a **P-N junction** semiconductor with a transparent window to allow light to reach the junction.
- **Characteristics**:
  - Operates in **reverse bias**.
  - Generates **photocurrent** when exposed to light.
  - Has a small **dark current** even without light.
  - **Responsivity** measures how effectively it converts light to current.
- **Working**: When light strikes the diode, it creates **electron-hole pairs** in the depletion region, generating a current.
- **Applications**: Used in **light detection**, **solar cells**, **camera sensors**, and **optical communication** systems.

Q6 Give symbol construction characteristics and working of light emitting diode

❖ Light Emitting Diode (LED):

- **Symbol**: Similar to a regular diode, with arrows pointing outward to indicate light emission.



- **Construction**: Made of a **P-N junction** semiconductor (e.g., **GaAs**, **GaN**) with a transparent cover to emit light.

- **Characteristics**:
  - Operates in **forward bias**.
  - Emits **light** when current flows through it.
  - **Efficient**: Converts electrical energy to light with minimal heat.
  - Light color depends on the **material** used.
- **Working**: When forward biased, electrons recombine with holes, releasing energy in the form of **light** (photons).
- **Applications**: Used in **displays**, **indicator lights**, **energy-efficient lighting**, and **optical communication**.

Q7 Explain need of rectifier in short

A **rectifier** is needed to **convert alternating current (AC) to direct current (DC)**. AC, which changes direction periodically, is not suitable for many electronic devices that require a steady, unidirectional flow of current (DC).

Need for a Rectifier:

1. **Power Supply for DC Devices**: Many electronic devices, such as computers, TVs, and batteries, require DC for operation, but most power sources provide AC. A rectifier converts AC to DC.

2. **Efficient Use of AC**: AC power is widely available (e.g., from power grids), but many applications (e.g., in electronics, motors) need DC for proper functionality.
3. **Energy Conversion**: Rectifiers are used in **power supplies**, **chargers**, and **solar panels** to convert and regulate power.

In short, rectifiers are crucial for providing DC from AC sources, enabling the use of AC power in devices that require DC.

Q8Explain half wave and full wave rectifier with diagram

Half Wave Rectifier

A half wave rectifier is a circuit that converts only one half of the AC (alternating current) cycle into DC (direct current). It uses a single diode to conduct during one half of the AC cycle and blocks the other half.

Working:

1. During the positive half-cycle of the AC input, the diode is forward-biased and allows current to flow through the load.

2. During the negative half-cycle, the diode is reverse-biased and blocks the current.

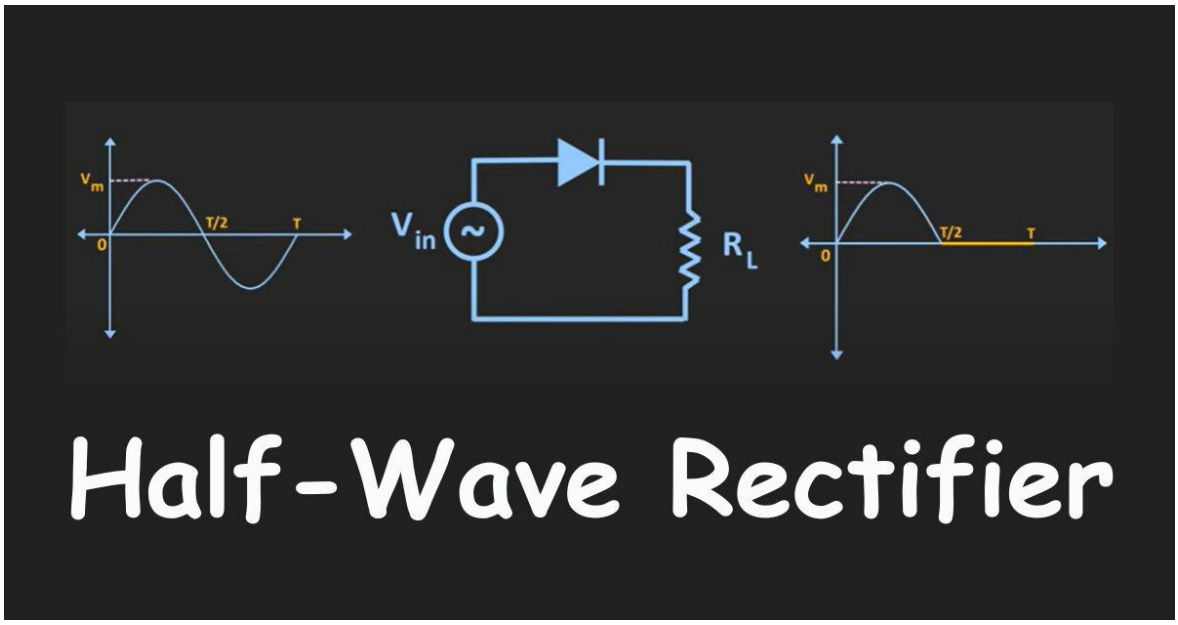
Characteristics:

Simpler design and fewer components.

Low efficiency (only one half of the waveform is utilized).

Ripple frequency is equal to the input AC frequency.

Diagram:



Full Wave Rectifier

A full wave rectifier converts both halves of the AC cycle into DC. It can be implemented using:

1. Center-Tapped Transformer with two diodes.

2. Bridge Rectifier with four diodes.

Working:

1. For the center-tapped design, during the positive half-cycle, one diode conducts while the other is reverse-biased, and vice versa for the negative half-cycle.

2. For the bridge rectifier, two diodes conduct during each half-cycle, ensuring current flows through the load in the same direction.

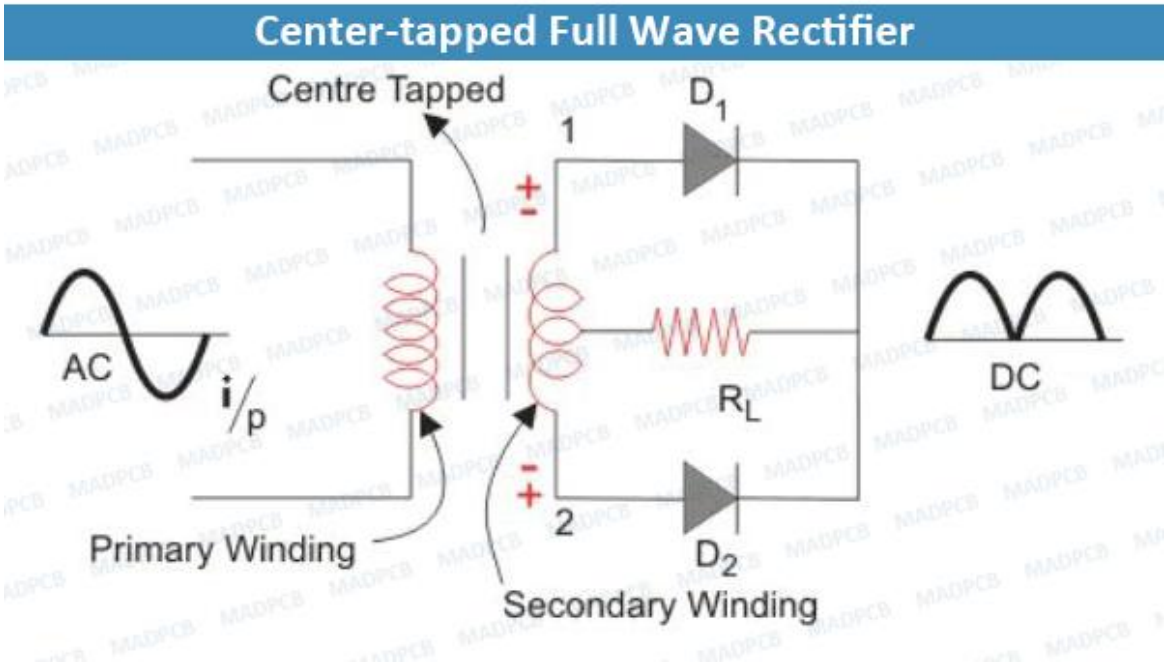
Characteristics:

Higher efficiency (utilizes both halves of the waveform).

Ripple frequency is double the input AC frequency.

Requires more components than the half-wave rectifier.

Diagram:


Center-tapped Full Wave Rectifier
Centre Tapped
D1
1
AC
ip
RL
DC
Primary Winding
2
D2
Secondary Winding